

Issue 1: September 2022

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# East Micronesia Cable System (EMCS)

## Maintenance Controller WebNSV

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Issue 5.0: November 2020

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NS Series

# WebNSV

**Element Management System  
Unified Management System**

Technical Description

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**NEC Corporation**

NEC Corporation Proprietary

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# 1 Introduction

NEC NS series WebNSV is a submarine network element management system compliant with the Telecommunication Management Network (TMN) specified in the ITU-T Recommendations.

The WebNSV provides all the functions required for submarine network management, which are Performance Management, Fault Management, Configuration Management, Security Management, and Wavelength Path Management.

The architecture of WebNSV is based on state-of-the-art Web and Java technologies that provide simple, optimized and easy-to-use management environment.

Figure 1 shows the overall configuration of submarine cable system with WebNSV management system.

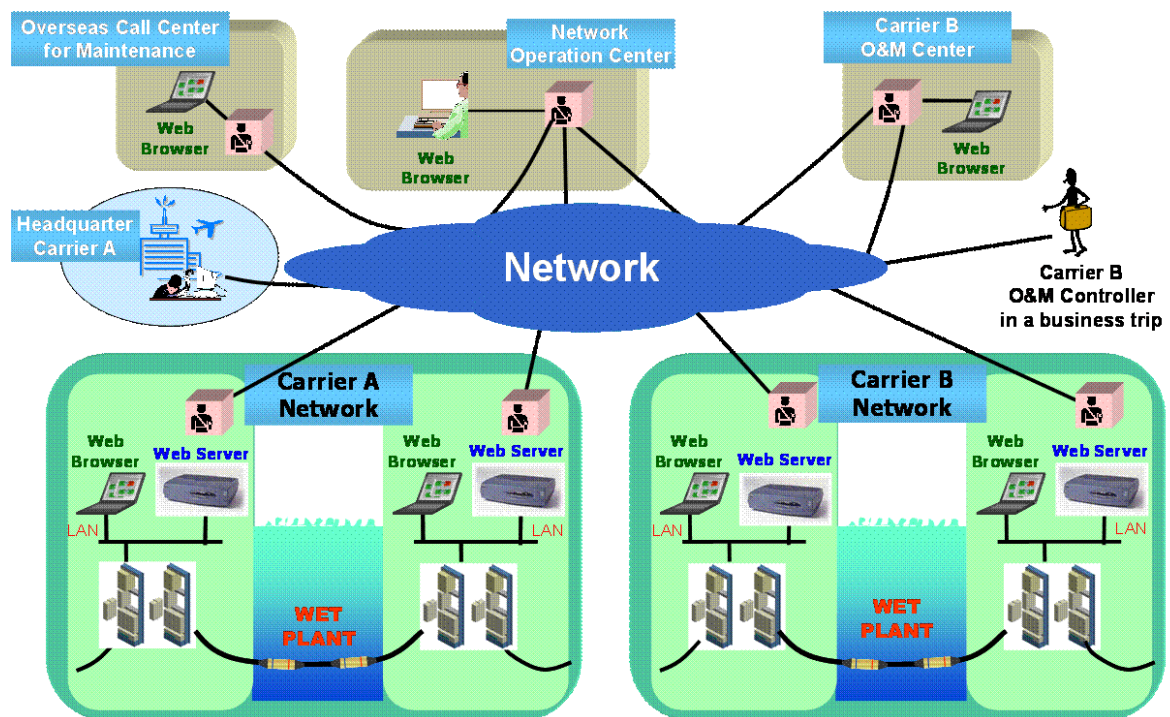


Figure 1 - Overall Submarine Cable System with WebNSV System Management Network

## 1.1 Architecture

The WebNSV consists of a network Element Management System (WebNSV EMS) and a Unified Management System (WebNSV UMS).

WebNSV EMS applies the distributed architectures that consists of the Web server (WebNSV EMS server and WebNSV UMS server) and Web browser (WebNSV client).

The WebNSV supports NEC's Line Terminal Equipment (LTE), Power Feed Equipment (PFE) and NEC's wet plant (Repeaters, Cable) through Remote Fiber Test Equipment (RFTE).

Figure 2 shows the structures for the WebNSV.

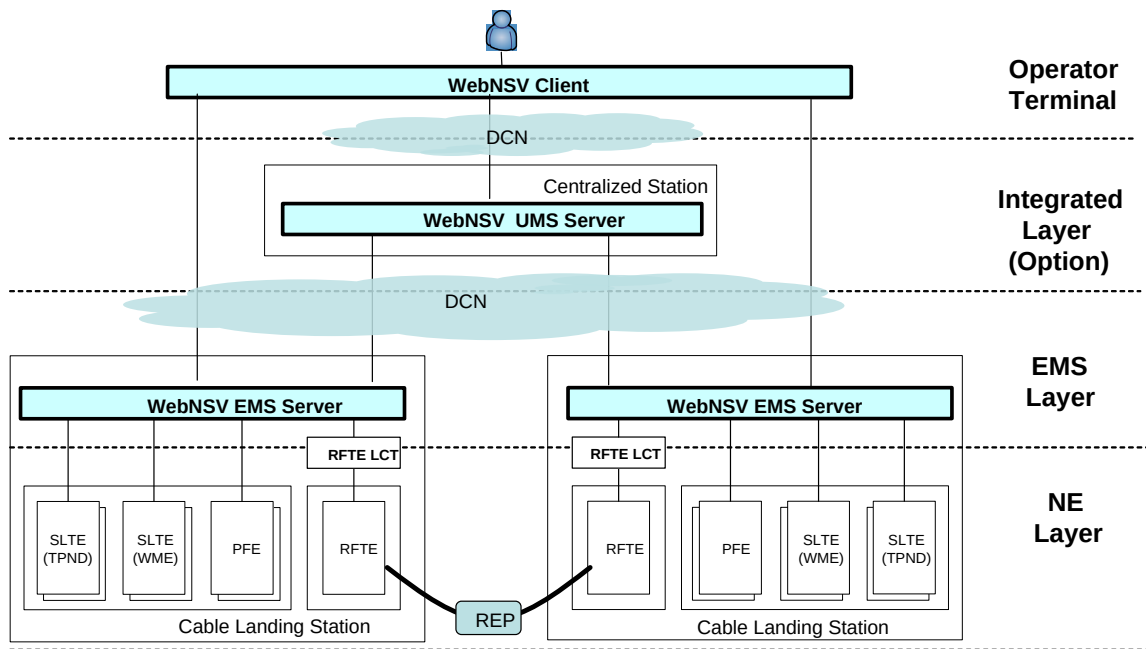


Figure 2 - WebNSV available structures (Sample)

The WebNSV EMS is installed in each cable landing station and supervises LTE, PFE and RFTE. All EMSs installed in different locations are connected through DCN, allowing for remote login to any WebNSV EMSs within the system from any stations.

The WebNSV UMS is a part of WebNSV system. It is an upper layer management system of the WebNSV EMS, and provides the integrated management functionality communicating with the WebNSV EMS at each cable landing station. It enables to monitor the entire system such as network powering, and the fiber/optical wavelength path among cable landing stations.

The WebNSV UMS is normally installed in the centralized station, which are linking to each cable station through DCN.

The WebNSV Client provides a Graphical User Interface (GUI) to monitor, the management data in the WebNSV EMS server and WebNSV UMS server.

The WebNSV Client can be used commonly for both types of servers as long as their communication links are available through DCN.

Figure 3 shows the available structures for the WebNSV UMS and EMS.

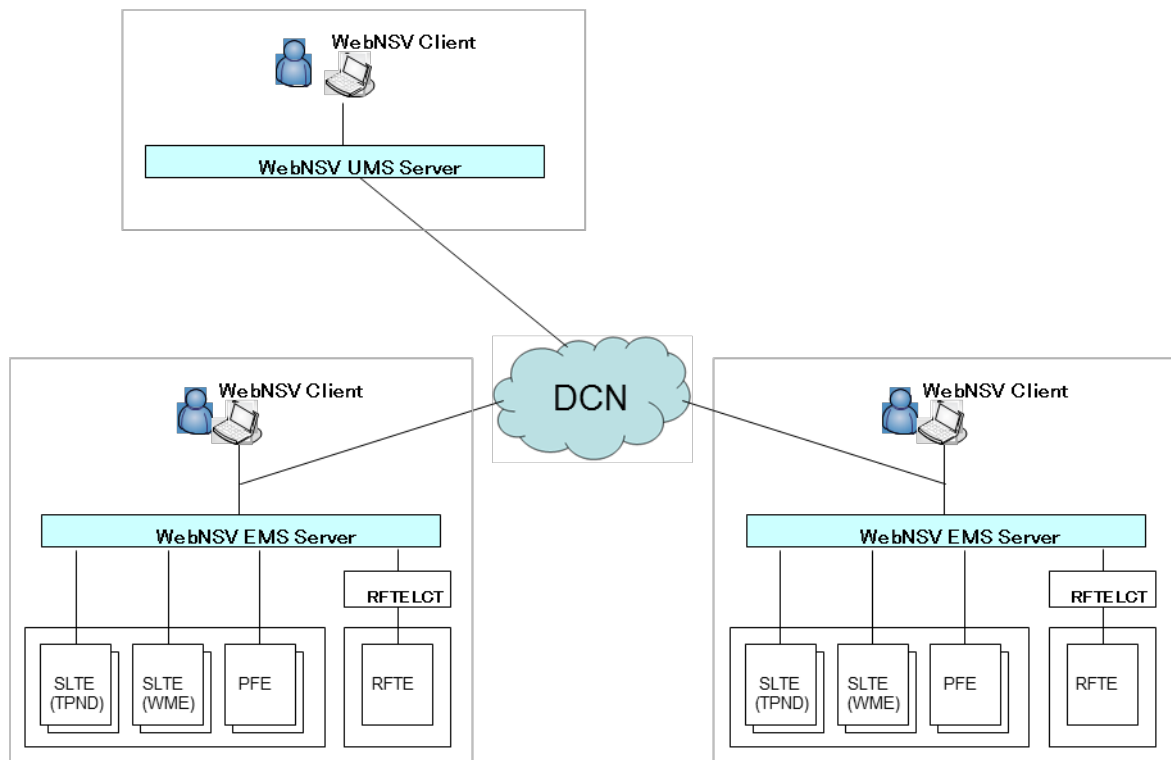


Figure 3 - WebNSV UMS and EMS available structures (Sample)

## 1.2 Key Features

The key features of the WebNSV UMS/EMS are summarized below.

- ✓ Real Time Monitoring  
The WebNSV UMS/EMS displays and updates current alarm, status and events in real time. No manual operations are required for operators to monitor the latest information.
- ✓ Location free  
The WebNSV UMS/EMS client terminal can be located at anywhere in the network: Landing Station, Network Operation Center, Remote Office etc.
- ✓ Step-by-step functional upgrades  
Additional functions can be added into the system after deployment. It enables operators to gradually upgrade its functions, and to reduce initial cost.
- ✓ Comprehensive and Customizable HMI  
Various Web HMI technologies such as hyperlink and applet are available. Drill-downable fault management achieved by hyperlink enables operators to the rapid diagnoses of the faulty conditions. HMI presentation is easily customizable according to operator's specifications.
- ✓ Sharing Functionality of Management information  
The WebNSV EMS basically manages NEs of local terminal station. Besides, each server has the sharing functionality of management information. Each server forwards its management information to other servers of each terminal station through the overhead channel of Transponder (TPND), and collects information from them.

## 2 EMS Management Functions

The management functions provided by the WebNSV EMS are summarized below. Each function details in the following sections.

- ✓ Fault Management
- ✓ Performance Management
- ✓ Configuration Management
- ✓ Log Management
- ✓ Security Management

## 2.1 Fault Management

The WebNSV EMS detects alarms from the managed network element in each cable station and notifies them to operators in visible and audible form.

The station symbols, NE symbols or other associated symbols on the management window of WebNSV Client changes colors upon receiving the alarm from the equipment. When the “Site View” is opened, the detail alarm journal is displayed together with the site configuration view. All of display provided for alarm notification is refreshed in real time.

All alarms detected by WebNSV EMS is recorded and stored. They are retrieved as the alarm history report.

[illegible]

Figure 4 - Top View (Sample)



## 2.2 Performance Management

The WebNSV EMS provides various capabilities, such as collection and reporting of alarms, as well as statistical analysis of performance data sent from NEs. The performance data includes amount of parity errors, FEC errors, unavailable time, etc. Those data are notified periodically from the respective NEs. The WebNSV EMS collects and stores the information in a database, and displays them upon operator's commands.

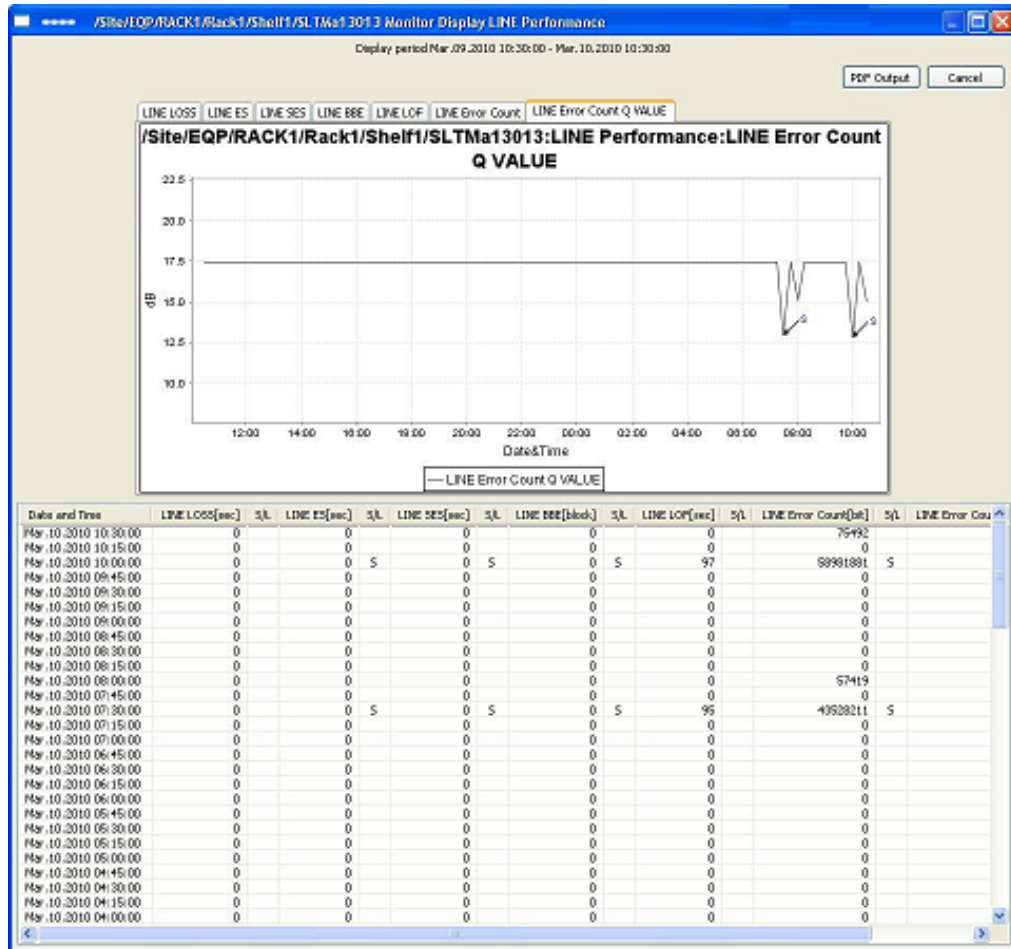


Figure 5 - Performance Monitoring (Sample)

## 2.3 Configuration Management

The Configuration Management provides the current equipment configuration and inventory functionalities. The current equipment configuration is automatically updated on in-service basis. The control functions are also provided for the LTE and the PFE. The WebNSV EMS restricts the access of unauthorized personnel by using operator's privilege levels in order to assure the system security. Furthermore, the EMS issues a notice for those operations before implementation.

## 2.4 Log Management

The Log Management provides the function of logging the event data and the performance data. The log data can be backed up to external device. The WebNSV EMS has functions to help diagnosing faulty events, such as filtering and printing of the log data.

**Log Display**

**Time Range**

| From  |     |      |      |        |        | To    |     |      |      |        |        |
|-------|-----|------|------|--------|--------|-------|-----|------|------|--------|--------|
| Month | Day | Year | Hour | Minute | Second | Month | Day | Year | Hour | Minute | Second |
| Jul   | 10  | 2007 | 15   | 53     | 52     | Jul   | 11  | 2007 | 15   | 53     | 52     |

NE Selection: ALL

Alarm: AVC StateChange Create / Delete Operation ALL

"Severity" Filter: 
☒ Cleared 
 ☒ Indeterminate 
 ☒ Minor 
 ☒ Major

"Probable Cause" Filter:

"Detail" Filter:

Operate:

| Severity | Time                 | Faulty Point | Proberble Cause | Detail     | Notification ID | Correlated ID |
|----------|----------------------|--------------|-----------------|------------|-----------------|---------------|
| Cleared  | Jan.11.2007 11:25:00 | user label   | cause           | detail     | 90              | 2             |
| Cleared  | Oct.02.2006 18:01:00 | faultyPoint  | probable        | additional | 2               | 90            |
|          |                      |              |                 |            |                 |               |
|          |                      |              |                 |            |                 |               |
|          |                      |              |                 |            |                 |               |
|          |                      |              |                 |            |                 |               |
|          |                      |              |                 |            |                 |               |
|          |                      |              |                 |            |                 |               |
|          |                      |              |                 |            |                 |               |
|          |                      |              |                 |            |                 |               |
|          |                      |              |                 |            |                 |               |
|          |                      |              |                 |            |                 |               |

page : 1 / 1      <<Back   Next>>

### Figure 6 - Terminal Log View (Sample)

## 2.5 Security Management

The Security Management provides access control for the WebNSV EMS. The WebNSV EMS can provide several different privilege levels to operators by using user ID and password.

The image shows a 'User Information' dialog box with a title bar and a close button. It contains three radio buttons for 'Operation': 'Add User' (selected), 'Delete User', and 'Modify User'. On the left is a 'Users' list box containing 'rootuser0', 'rootuser4', and 'root'. The main area is divided into three sections: 'User Profile' with fields for 'User Name', 'Password', 'Retype Password', and a 'User Level' dropdown set to '1'; 'IP Address Filter' with four dropdown menus for octets, 'Add', and 'Delete' buttons, and a text area; and 'MAC Address Filter' with six dropdown menus for octets, 'Add', and 'Delete' buttons, and a text area. At the bottom right are 'Add User' and 'Cancel' buttons.

Figure 7 - User Entry View (Sample)

### 3 UMS Management Functions (Option)

The WebNSV UMS provides the integrated management functions communicating with WebNSV EMSs. All functionalities provided by WebNSV EMS are available from the WebNSV UMS by invoking the WebNSV EMS applications.

The followings are the dedicated management functions provided by the WebNSV UMS.

- ✓ Integrated Fault Management
- ✓ Log Management
- ✓ DWA Path Management
- ✓ Power Feeding Management
- ✓ Fiber Path management
- ✓ Security Management

#### 3.1 Integrated Fault Management

The WebNSV UMS provides the integrated fault management functions, which enables to monitor all of the stations in the system at the same time. When the WebNSV EMSs in any stations detects alarms from managed equipments, these alarms are displayed on the “Top View” on the WebNSV UMS.

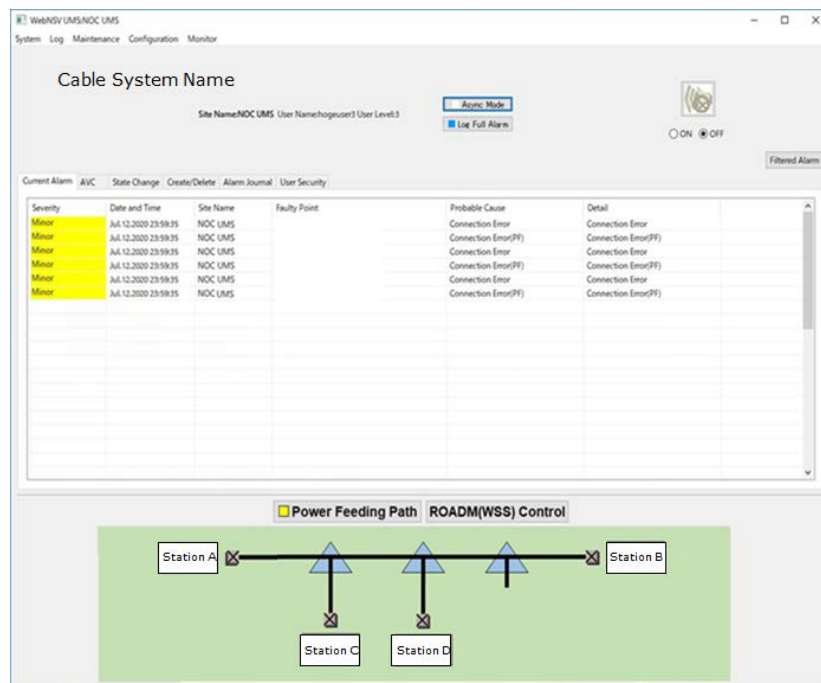


Figure 8 – Top View (Sample)

### 3.2 Log Management

The Log Management provides the function of logging the event data for all stations. The log data can be backed up to external device. The WebNSV UMS has functions to help diagnosing faulty events, such as filtering and printing of the log data.

The screenshot shows a 'Log Display' window with a search and filter interface at the top. Below the filters is a table of log entries. The table has the following columns: Severity, Time, Faulty Point, Probable Cause, Detail, Notification ID, and Correlated ID. The first two rows of data are visible.

| Severity | Time                 | Faulty Point | Probable Cause | Detail     | Notification ID | Correlated ID |
|----------|----------------------|--------------|----------------|------------|-----------------|---------------|
| Alarm    | Jan.11.2007 11:25:00 | user label   | cause          | detail     | 90              | 2             |
| Cleared  | Oct.02.2006 18:01:00 | faultyPoint  | probable       | additional | 2               | 90            |

At the bottom right of the window, it says 'page : 1 / 1' and has '<<Back' and 'Next>>' buttons.

Figure 9 – Terminal Log View (Sample)

### 3.3 DWA Path Management

The WebNSV UMS provides the DWA Path Management function, which enables to manage the DWA path connection and monitor their status for the direct wavelength access application.

By clicking the cable line on the “Top View” on the WebNSV UMS, the detail of DWA path connection information (DWA Path View) are available. The DWA path on the WebNSV UMS changes the color upon the associated alarm is notified from the WebNSV EMS.

The DWA Path Management provides functions as follows.

- ✓ Configuration Management
- ✓ Fault Management
- ✓ Performance Management
- ✓ Log Management
- ✓ Security Management

Above functions are provided by GUI which shows DWA path connection, alarm status and performance data etc. for each DWA path. It supports operator to create/delete DWA paths in accordance with the physical path connection between the client interface of LTES. Figure 10 shows an example of the DWA Path View

The operator is able to assign a unique name to each DWA path, monitor and print the alarm and/or PM records with filtering function.

Figure 11 shows an example of the DWA Path Setting Window

The DWA Path Setting window is called from the DWA Path View, and the operator is able to register the DWA Path.

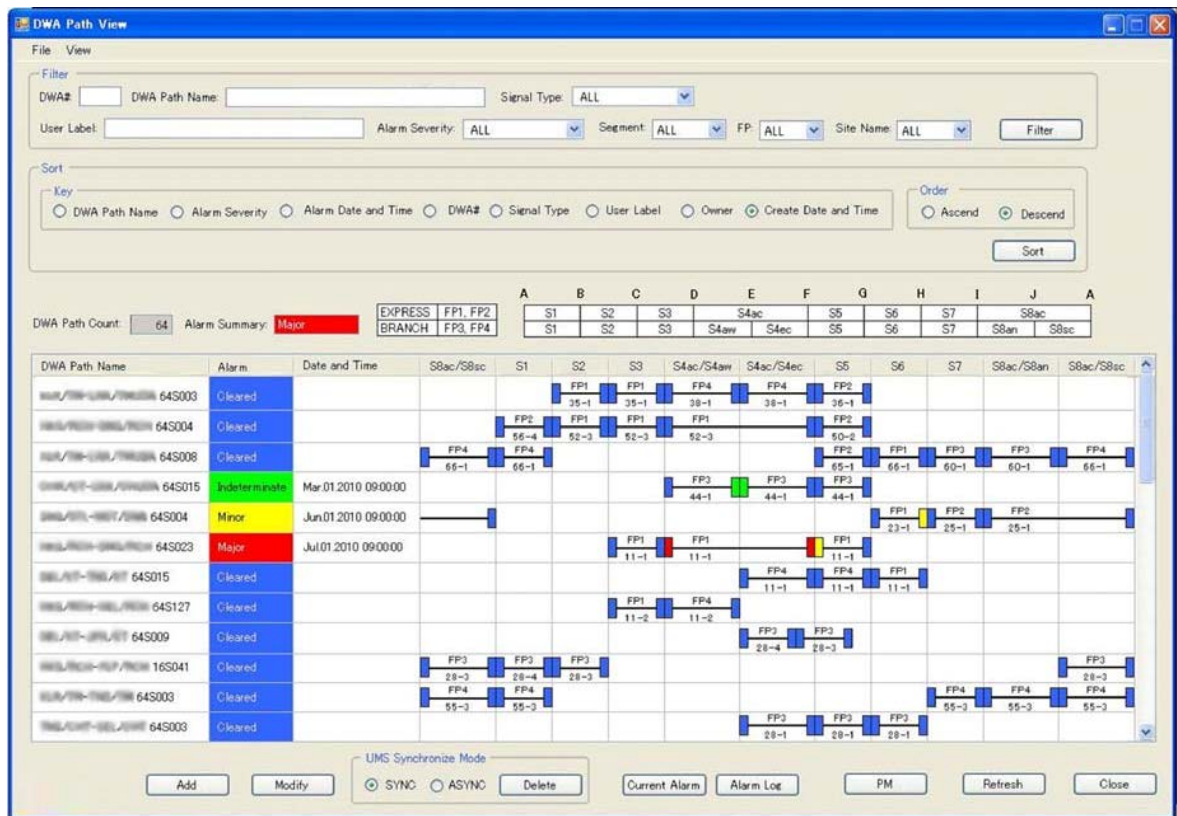


Figure 10 – DWA Path View (Sample)

The DWA Path Setting window allows configuration for stations A through J. Each station has settings for FP (Function Point) and CH (Channel). The UMS Synchronize Mode is set to SYNC.

Station A: S1, S2, S3, S4ac, S4aw, S4ec, S5, S6, S7, S8ac, S8an, S8sc

Station B: S1, S2, S3, S4ac, S4aw, S4ec, S5, S6, S7, S8ac, S8an, S8sc

Station C: S1, S2, S3, S4ac, S4aw, S4ec, S5, S6, S7, S8ac, S8an, S8sc

Station D: S1, S2, S3, S4ac, S4aw, S4ec, S5, S6, S7, S8ac, S8an, S8sc

Station E: S1, S2, S3, S4ac, S4aw, S4ec, S5, S6, S7, S8ac, S8an, S8sc

Station F: S1, S2, S3, S4ac, S4aw, S4ec, S5, S6, S7, S8ac, S8an, S8sc

Station G: S1, S2, S3, S4ac, S4aw, S4ec, S5, S6, S7, S8ac, S8an, S8sc

Station H: S1, S2, S3, S4ac, S4aw, S4ec, S5, S6, S7, S8ac, S8an, S8sc

Station I: S1, S2, S3, S4ac, S4aw, S4ec, S5, S6, S7, S8ac, S8an, S8sc

Station J: S1, S2, S3, S4ac, S4aw, S4ec, S5, S6, S7, S8ac, S8an, S8sc

Figure 11 – DWA Path Setting Window (Sample)



Figure 12 shows an example of the DWA path Current Alarm Window  
The Current Alarm window is the window displaying the DWA Current Alarm list.

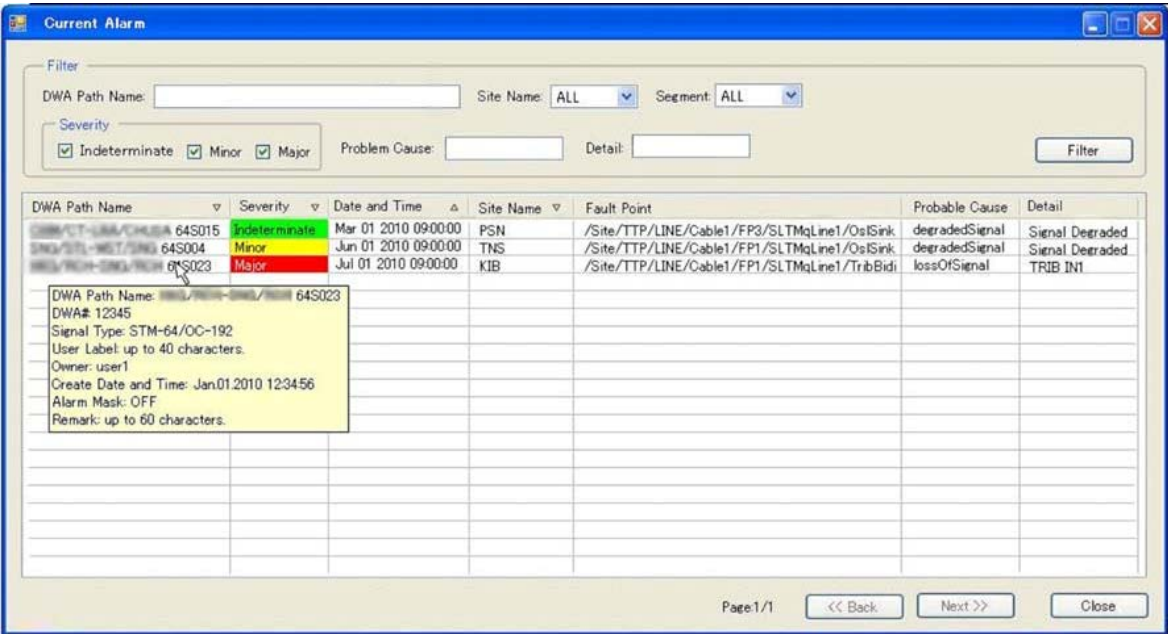


Figure 12 – DWA path Current Alarm Window (Sample)

Figure 13 shows an example of the DWA path Alarm Log Window.  
The Alarm Log window is the window displaying the DWA Path Alarm history.

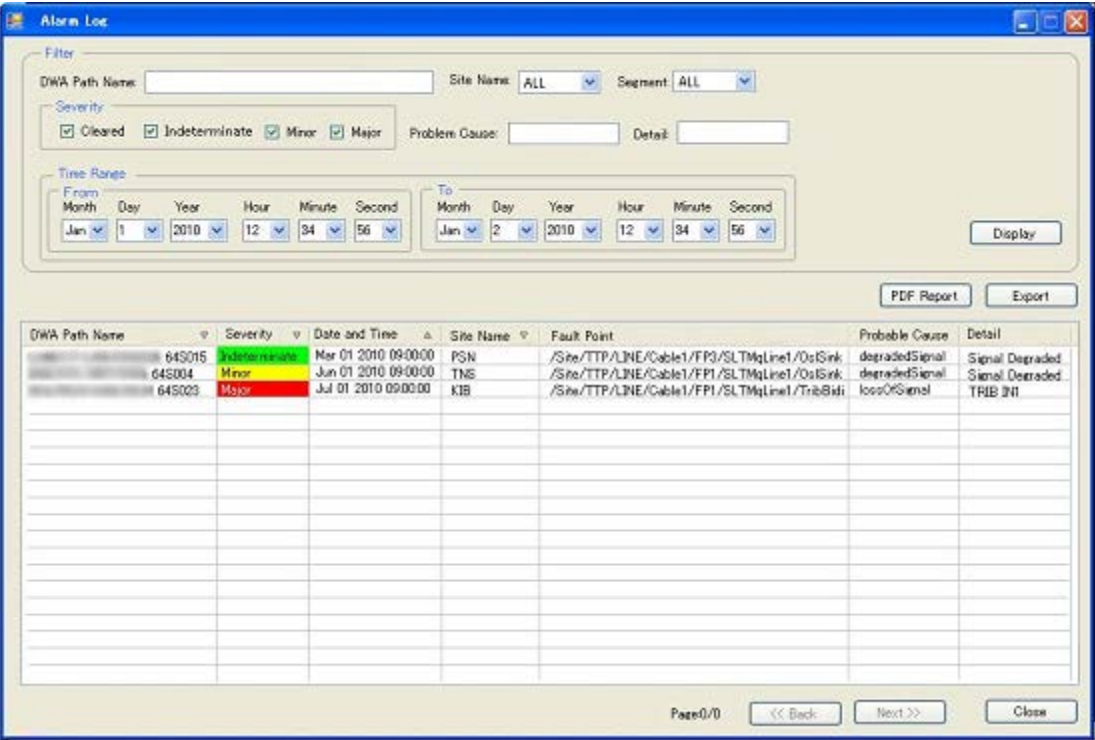


Figure 13 –DWA path Alarm Log Window (Sample)



Figure 14 and Figure 15 show an example of the DWA path Monitor Display Window.

The Monitor Display window is the window which specifies the DWA Path monitoring information display conditions.

Figure 14 – DWA path Monitor Display Window (Sample)

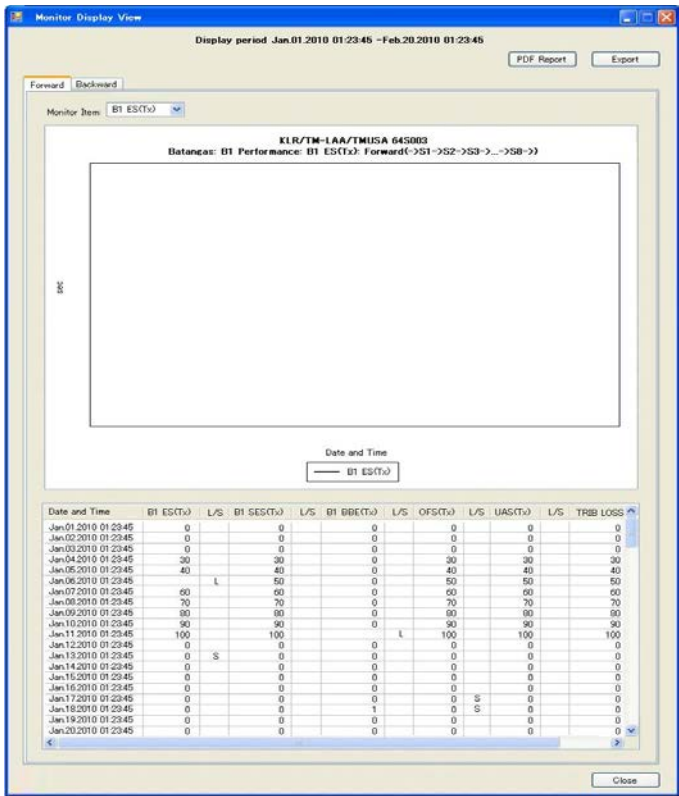


Figure 15 – DWA path Monitor Display Window (Sample)

### 3.4 Power Feeding Management

The WebNSV UMS provides the power feeding path management function, which enables to monitor the system powering status and PFE's operating conditions (current, voltage, operated regulation mode) in the entire system.

For the system deployed with the remotely operational Power Switching Branching Unit (PSBU), the BU status and controlling is managed by this function.

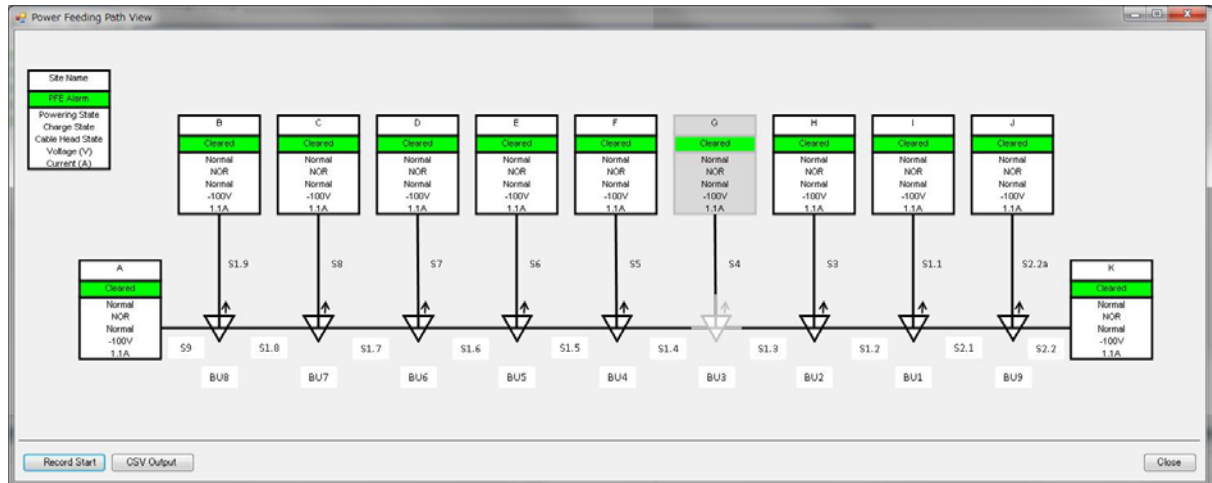


Figure 16 – Power Path View (Sample)

### 3.5 Fiber Path Management

The WebNSV UMS provides the Fiber Path management function, which enables to manage the fiber path connection and monitor their status.

By clicking the cable line on the “Top View” on the WebNSV UMS, the detail of fiber connection information (Fiber Path View) is available. The optical fiber on the WebNSV UMS changes the color upon the associated alarm is notified from the WebNSV

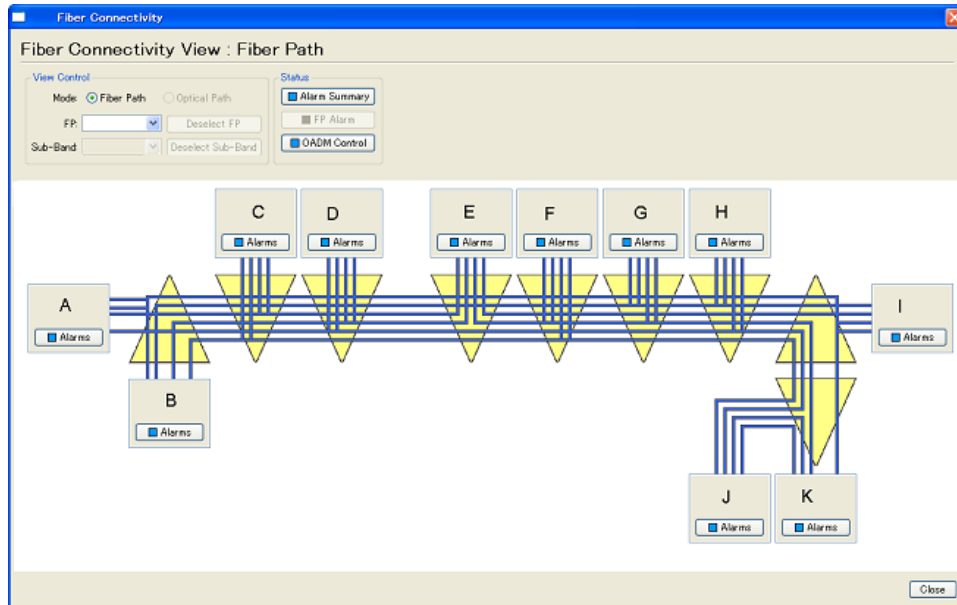


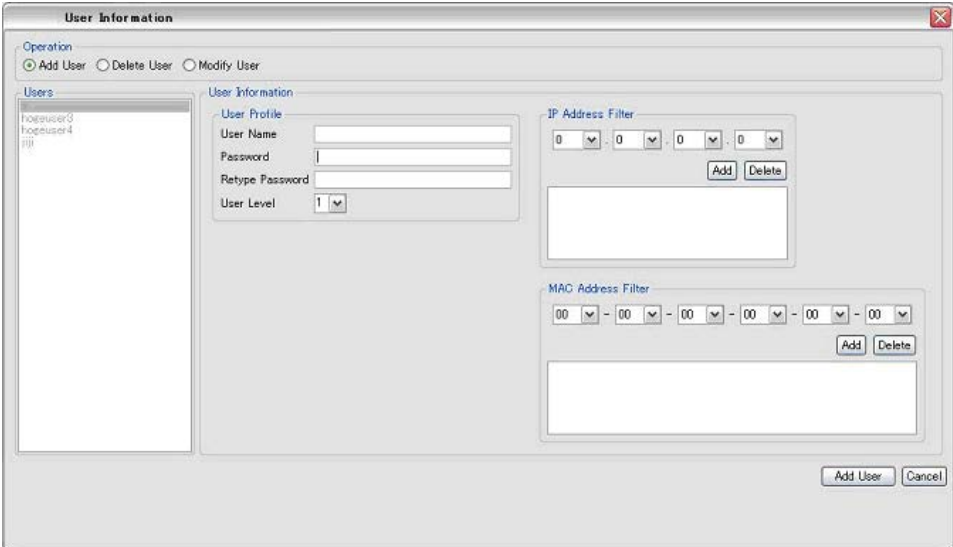
Figure 17 – Fiber Path View (Sample)

### 3.6 ROADM BU Management

The WebNSV UMS provides the ROADM BU management function, which enables to manage variable add/drop bandwidth of ROADM BU.

### 3.7 Security Management

The Security Management provides access control for the WebNSV UMS. The WebNSV UMS can provide several different privilege levels to operators by using user ID and password.



The image shows a 'User Information' dialog box with a title bar and a close button. It contains three radio buttons for 'Operation': 'Add User' (selected), 'Delete User', and 'Modify User'. On the left is a 'Users' list box containing 'hoesuser0', 'hoesuser4', and 'root'. The main area is titled 'User Information' and contains a 'User Profile' section with fields for 'User Name', 'Password', 'Retype Password', and a 'User Level' dropdown set to '1'. To the right are two filter sections: 'IP Address Filter' and 'MAC Address Filter', each with a row of six dropdown menus and 'Add'/'Delete' buttons. At the bottom right are 'Add User' and 'Cancel' buttons.

Figure 18 – User Entry View (Sample)

## 4 Software and Hardware

### 4.1 Software Architecture

The WebNSV UMS and WebNSV EMS software are composed of the following modules. The applied platforms and applications are decided depending on network scale.

The system applications and middlewares installed in WebNSV EMS Server, and/or WebNSV UMS Server are independent from the vision of hardware or operation system. Therefore the major softwares modifications are unnecessary when the hardware upgrade is needed in the futures.

#### WebNSV UMS Server and WebNSV EMS Server

- Server OS: Linux
- WEB Server: Apache
- Database: PostgreSQL

#### WebNSV Client

- Client OS: Windows
- WEB browser: Internet Explorer
- System Applications

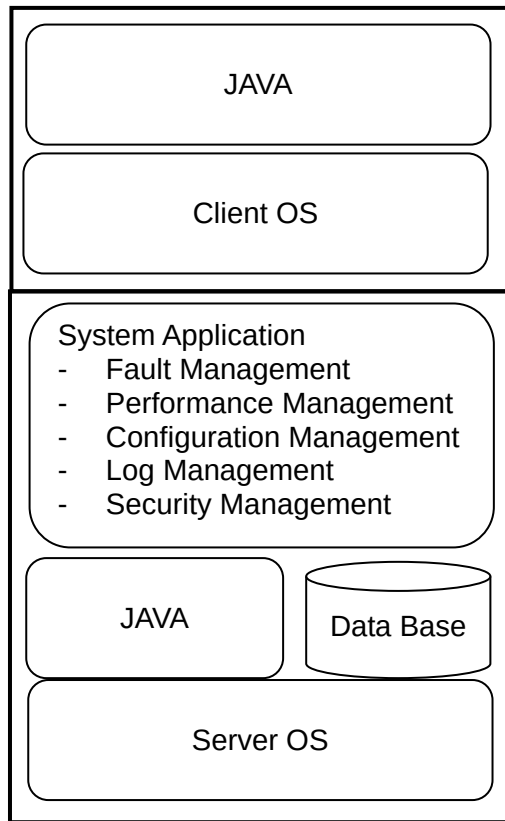


Figure 19 – WebNSV EMS / WebNSV UMS Software Architecture

## 4.2 Hardware Specifications

The general hardware specifications required for the WebNSV UMS/EMS server and client are as follows.

### WebNSV UMS Server and WebNSV EMS Server

- CPU: Intel Xeon Processor 1.5GHz or more
- HDD: 40G or more
- Memory: 2GB or more
- LAN interface: 2port 10/100BASE-T (RJ45)
- AC voltage AC100-240V
- Cluster model is available as a hot-standby server configuration upon the Purchaser's requests.

### WebNSV Client

- CPU: Intel Pentium 4 Processor 1GHz or more
- HDD: 20G or more
- Memory: 512MB or more
- LAN interface: 1port 10/100BASE-T (RJ45)
- AC voltage AC100-240V

## 5 Management Network

This section describes the management network architecture for the WebNSV System. Figure 20 and Figure 21 show the typical architecture.

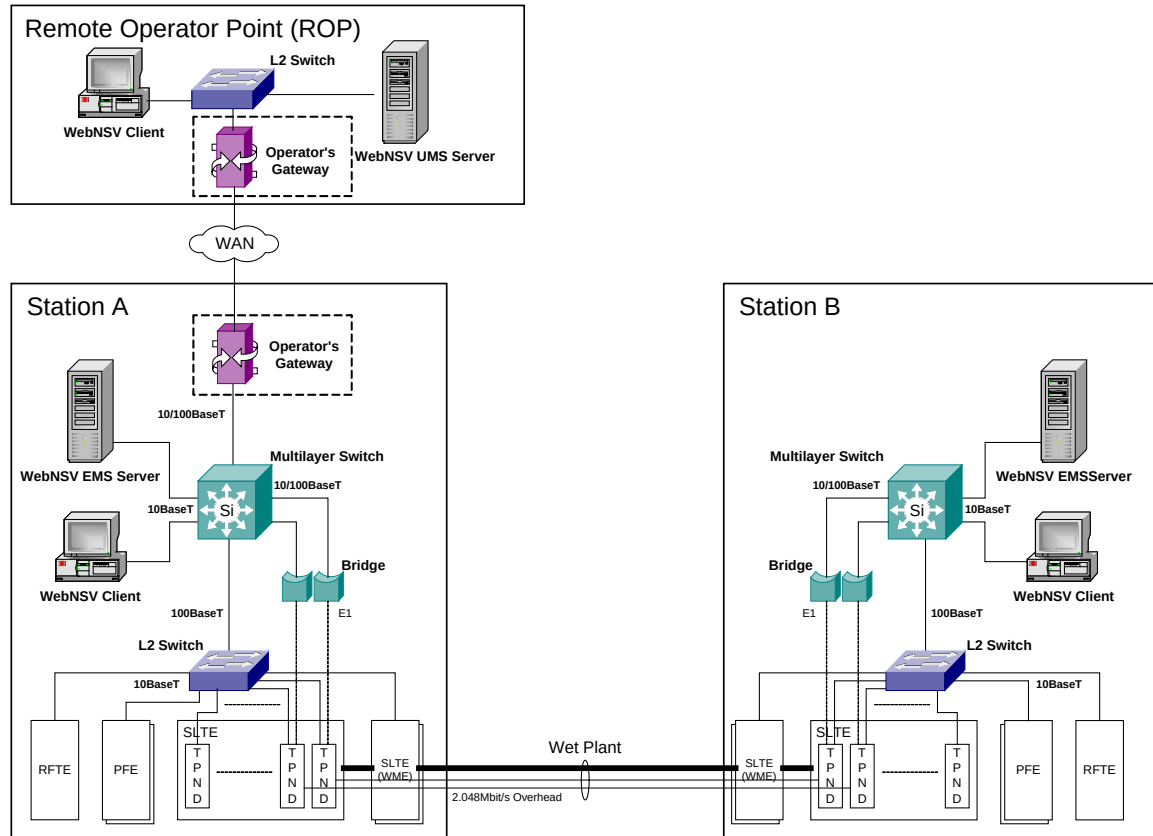


Figure 20 – Typical Management Network Architecture (Sample for medium-size network)



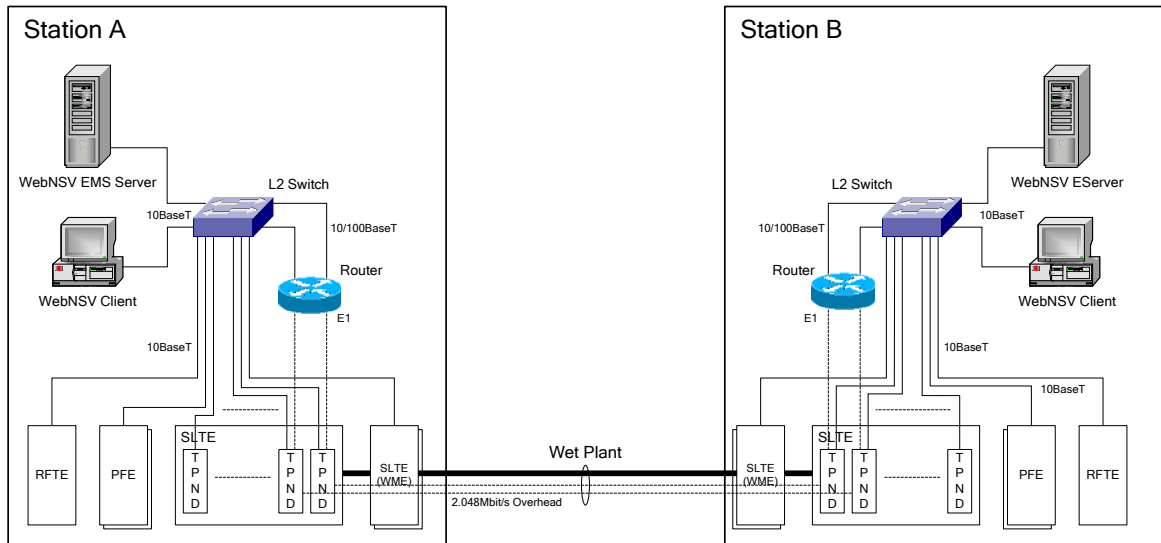


Figure 21 – Typical Management Network Architecture (Sample for small-size network) without WebNSV UMS

## 5.1 Inter-station Communication Network

The inter-station communication network is required for the server to server communication. By using the 2.048Mbit/s overhead channels available in NEC's Submarine Line Terminating Equipment, the management information of the WebNSV server is carried to other stations. The overhead channels are terminated at TPND, and converted to LAN interfaces by Data Communication Network (DCN) equipment such as a bridge shown in Figure 20 or a router shown in Figure 21. Generally, 2 overhead channels are used for this communication for redundancy.

## 5.2 In-station Communication Network

The in-station communication network is entirely configured with LAN interfaces based on TCP/IP transport protocol. This network is required for making the connectivity between the EMS server and Network Elements (NEs), between WebNSV EMS and UMS servers, and between the server and client. The WebNSV EMS and UMS server, WebNSV client and each NE provides 10BASE-T interfaces, and communicates each other through DCN equipment.

The required DCN equipment depends on a system configuration and a network scale. In case of Figure 20, L2 switch and Multi Layer Switch (MLS) are applied, assuming a medium-size network composed of several terminal stations with the connectivity of a Remote Operator Point (ROP). The L2 switch aggregates LAN interfaces from Each Network Element (NE), and interfaces with MLS using 100BASE-T interface. MLS provides LAN interfaces to the WebNSV EMS server, client and other peripheral equipment. Besides, MLS provides the terminating and IP routing functions for WAN from other terminal stations and a ROP.

### 5.3 Connectivity with ROP

The WebNSV can provide the remote monitoring function by using operator's network (WAN). The WebNSV can make connectivity with a Remote Operator Point (ROP) via operator's gateway terminating WAN network by using 10 or 100BASE-T interface. The required bandwidth between a terminal station and a remote station is 2Mbit/s or higher.

### 5.4 Data Communication Network Equipment

In order to configure the management network, several DCN equipments are required. Although Figure 20 and Figure 21 show typical configurations, the required DCN equipment are decided depending on a system configuration and a network scale. This section depicts the general requirements and specifications.

- ✓ For connectivity with other terminal stations

For the connection among terminal stations, the 2.048Mbit/s overhead channels in the NEC's SLTE are available. Applied DCN equipment should be able to terminate the overhead channels and convert to LAN interfaces. Moreover, the DCN equipment should be equipped with a general IP routing function to realize dynamic routing and interconnect networks with different sub-net masks.

- ✓ For connectivity inside a terminal station

At a terminal station, L2 switch is required in order to terminate LAN interfaces from NEs, the WebNSV EMS and/or peripheral equipment.

### 5.5 Network Time Stamp

WebNSV EMS in each cable station set the time of its managed NE. This time set is done at the time when the WebEMS and NE connection is established and also once a day periodically.

In order to achieve entire system time accuracy, Network Time Protocol (NTP) server is recommended to be installed at one of stations within the network. The WebNSV UMS server and EMS servers are slaved to the NTP server via DCN.

Note: This document describes the current version of NEC standard equipment. If there is any conflict between this document and the system description and/or the compliance statement, the latter will supersede this document. The specifications or configuration contained in this document are subject to change without notice due to NEC's continuing design improvement.